

Master of Science in Precision Scientific Instrumentation (MSPSI)

Guidelines for MSPSI Dissertations

FORMAT

1. The dissertation should be typewritten in English, in the third person and with one and a half spacing between lines. There is no prescribed length for the dissertation; a good guide would be around 60 to 100 pages printed in font size 12, with double-sided printing. You are strongly encouraged to use your own words instead of copying directly from books, technical reports, generative AI, etc. Standard International Units (SI) should be used.
2. White A-4 sized bond paper of at least 80gsm should be used (applicable for hard copy of the dissertation when requested by the examiner).
3. The margin on the left-hand side must be about 3.5 cm. On the top, bottom and right-hand side, a margin of 3 cm is recommended.
4. The same type setting should be used in the main text of the report to ensure that the letters do not vary in size, type and shade. One and a half line spacing should be used for the text of the report.

Below is the summary table:

Component	Assessor
Page	White A4 size bond paper of at least 80 g/m ² (applicable for hard copy of the dissertation when requested by examiner).
Text	Times New Roman, size: 12
Margins	3.5cm margin on the left-hand side and 3cm on the top, bottom and right-hand sides of each page.
Typing	The same font and pitch for the whole report except when highlighting important matters
Spacing	One-and-a-half line spacing
Length	60 to 100 printed pages, preferably printed double sided (applicable for hard copy of the dissertation when requested by examiner).

5. The contents of the report should be presented in the following order:
- (1) Title page
 - (2) Statement of Originality
 - (3) Supervisor Declaration Statement
 - (4) Authorship Attribution Statement
 - (5) Table of contents
 - (6) Abstract
 - (7) Acknowledgment page to give recognition of any advisory or financial assistance received in the course of the work on which the report is based
(optional)
 - (8) Acronyms *(optional)*
 - (9) Symbols *(optional)*
 - (10) List of figures
 - (11) List of tables
 - (12) Introductory chapter
 - (13) Text chapters
 - (14) References
 - (15) Appendix/appendices *(optional)*
6. A soft copy of dissertation, Turnitin Originality Report (in HTML format) must upload to the Dissertation Submission E-Form for submission.
7. After the dissertation has been examined and approved, students must submit a soft copy of the final version of the dissertation to the School through his supervisor and upload a copy to NTU Library (DR-NTU).
8. Samples of the cover page, title page, 3 declaration statements and Dissertation format, as well as an extract on the method of listing references, are attached in Appendices 1 to 6 respectively.

STANDARD OF WRITING

The dissertation should be logically laid out. The writing should be in grammatically correct, clear and concise English. There are many books in the NTU library, providing guidance on writing technical reports.

Any attempt to provide guidance or feature common errors here will run into many pages. So, we will refrain from doing that except to say one thing: if a word (or phrase

or sentence) can be deleted without changing the meaning of what you want to say, then delete it.

All figures and tables should be numbered sequentially, chapter by chapter, and be given a caption. Each must be referred to in the main text, and wherever possible appear near to where it is referred to.

CONTENTS

First and foremost, the contents of your work must be relevant to the MSc Program you are pursuing. Rather than the actual contents, which will vary from dissertation to dissertation and from program to program, this section describes the general areas, which a dissertation should address. While these areas are not necessarily universal, i.e., common to all dissertations, they are more the norm than the exception. The sample Contents page in Appendix 6 gives a pretty good summary of these areas.

1. The Abstract should be a short and concise passage on the important work and contributions of the project: the motivation and the problem pursued, the method you employed and the results obtained, highlighting the significant achievements. It should not contain peripheral things like summary of literature review, and it is not good enough to say that a certain issue has been studied without stating the results of the study. Generally, one page is about the right length for the Abstract.
2. The first chapter of the dissertation is almost invariably the Introduction. Generally, its purpose is to lead the readers into the problem you intend to attack in the project, to set the scene. The main points here consist of the background to the problem and your motivation in solving it. This then leads to the objectives and the scope of the project. It is good to conclude your Introduction with a section on the layout of the dissertation. It prepares the readers for what is to come.
3. Then comes the main part of your work. To lay the ground, there should first be a chapter on what has been done before on the problem - a Literature Review. This is an important section because it shows that you do not narrowly focus only on what you do, but are aware of the work related elsewhere, some of which might be instructive to your solving the problem. It can also explain why you are taking the direction you do.
4. The next few chapters should describe the work you have done in tackling the problem. There might be a chapter on the fundamental theories relevant to the solution you are pursuing, or the supporting technologies you need in implementing

the solution. Then there should be a chapter on the solution itself, followed by a chapter on the results and analysis of the results.

5. The last chapter is always the Conclusion. This generally should have three parts. The first is a concise summary of the work you have done. In a way, this is like the abstract. Then there is the conclusion, in which you highlight the significance of the results, and perhaps the consequences of the results, critically where necessary. The last thing is usually recommendations and/or future work, in which you identify the inadequacies of what you have done, and suggest how the gaps may be plugged.
6. Generally, there should be no more than six or seven chapters in your dissertation. If you have more than that, you should take a close look at its organization and see if certain chapters can be merged.

SUPPLEMENTARY ELEMENTS

1. Documents that are prepared with the help of other sources should have a list of sources cited. A list of **References** contains only sources the writer quotes directly, takes original ideas from, and refers to in the dissertation should be included. In reports where the subject is primarily scientific, the list of references is the most widely accepted way to cite specific sources.
2. The **Appendix** contains related data not necessary to the immediate understanding of the discussion in the report. This may contain materials such as: tables, graphs, illustrations, description of equipment, samples of forms, data sheets, questionnaires, equations, and any material that must be included for record purposes.

Each entry (sample forms, detailed data for references, tables, pictures, questionnaires, charts, maps, graphic representations) in the appendix requires an identifying title. Every entry in the appendix must be referred to in the body of the report. Each appendix must be lettered, beginning with Appendix A. The list of appendices should be appearing in the table of contents following the list of references entry.

APPENDIX 1: COVER PAGE



**NANYANG
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SINGAPORE

[DISSERTATION TITLE]

[AUTHOR'S NAME]

SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES

[SUBMISSION YEAR, e.g. 2026]

APPENDIX 2: TITLE PAGE

[DISSERTATION TITLE]

[AUTHOR'S NAME]

SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES

A DISSERTATION SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE DEGREE OF

**[PROGRAM NAME, e.g. MASTER OF SCIENCE IN PRECISION
SCIENTIFIC INSTRUMENTATION]**

[SUBMISSION YEAR, e.g. 2026]

APPENDIX 3: STATEMENT OF ORIGINALITY

I hereby certify that the work embodies in this dissertation is the result of original research, is free of plagiarized materials, and has not been submitted for a higher degree to any other University or Institution.

[Input Date here]

Date

[Sign here]

[Student Name] & Signature

APPENDIX 4: SUPERVISOR DECLARATION STATEMENT

I have reviewed the content and presentation style of this Dissertation. I declare that it is free of plagiarism and of sufficient grammatical clarity to be examined. To the best of my knowledge, the research and writing are those of the candidate except as acknowledged in the Authorship Attribution Statement. I confirm that the investigations were conducted in accord with the ethics policies and integrity standards of Nanyang Technological University and that the research data are presented honestly and without prejudice.

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[Supervisor Name] & Signature

APPENDIX 5: AUTHORSHIP ATTRIBUTION STATEMENT

Please select one of the following: *delete as appropriate:

*(A) This thesis **does not** contain any materials from papers published in peer-reviewed journals or from papers accepted at conferences in which I am listed as an author.

*(B) This thesis contains material from **[x number]** paper(s) published in the following peer-reviewed journal(s) / from papers accepted at conferences in which I am listed as an author.

Please amend the typical statements below to suit your circumstances if (B) is selected.

Chapter 4 is published as D.T. Murphy, S. Schmid, J.R. Hester, P.E.R. Blanchard, and W. Miiller. Coordination site disorder in spinel-type LiMnTiO₄. *Inorganic Chemistry* **54**, 4636-4643 (2015). DOI: 10.1021/ic502747p.

The contributions of the co-authors are as follows:

- A/Prof Schmid provided the initial project direction and edited the manuscript drafts.
- I prepared the manuscript drafts. The manuscript was revised by Dr Hester and Dr. Blanchard.
- I co-designed the study with A/Prof Siegbert Schmid and performed all the laboratory work at the School of Materials Science and Engineering and the Singapore Synchrotron Light Source. I also analyzed the data.
- All microscopy, including sample preparation, was conducted by me in the Facility for Analysis, Characterization, Testing and Simulation.
- Dr James Hester assisted in the collection of the neutron powder diffraction data.
- Dr Peter Blanchard assisted in the interpretation of the X-ray absorption spectroscopy data and carried out the spectral interpretation.
- Dr Wojciech Miiller assisted in the collection and provide guidance in the interpretation of the magnetic measurement data.

Chapter 5 is published as H. V Doan, B. Yao, Y. Fang, A. Sartbaeva, U. Hintermair, V. P Ting, Controlled Formation of Hierarchical Metal-Organic Frameworks using CO₂ Expanded Solvent Systems. In press, *ACS Sustainable Chemistry & Engineering* (2017). DOI: 10.1021/acssuschemeng.7b01429

The contributions of the co-authors are as follows:

- Prof Ting suggested the materials area and edited the manuscript drafts.
- I wrote the drafts of the manuscript. The manuscript was revised together with Dr. Sartbaeva and Dr. Yao.
- I performed all the materials synthesis, collected X-ray diffraction patterns and visible light spectra, carried transmission electron microscopy, and conducted data evaluation.
- Dr. Y. Fang conducted the Rietveld analysis of the powder X-ray diffraction data and single crystal structure determinations.
- Dr U. Hintermair conducted the molecular dynamics simulations.
- Ms. A. Sartbaeva prepared the samples for electron microscopy.

[Input Date here]

Date

[Sign here]

[Student Name] & Signature

APPENDIX 6: DISSERTATION FORMAT

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Abstract

[Insert Abstract here]

Acknowledgements (*Optional*)

[Insert Acknowledgements here]

List of Abbreviations (*Optional*)

AFM	Atomic Force Microscope / Microscopy
AMR	Anisotropic Magnetoresistance
BPM	Beats Per Minute
[Insert Acronym]	[here]

List of Symbols *(Optional)*

[Insert Symbol]	[here]
[Insert Symbol]	[here]
[Insert Symbol]	[here]
[Insert Symbol]	[here]

List of Figures

List of Tables

Chapter 1

Introduction

In this Dissertation... [Insert Introduction here]

1.1 Motivations

1.2 Objectives and Scope

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6.1 Conclusions

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Appendix *(Optional)*